



Signals and Systems

L-Transform

(lecture 34, Date:-27/04/2026)

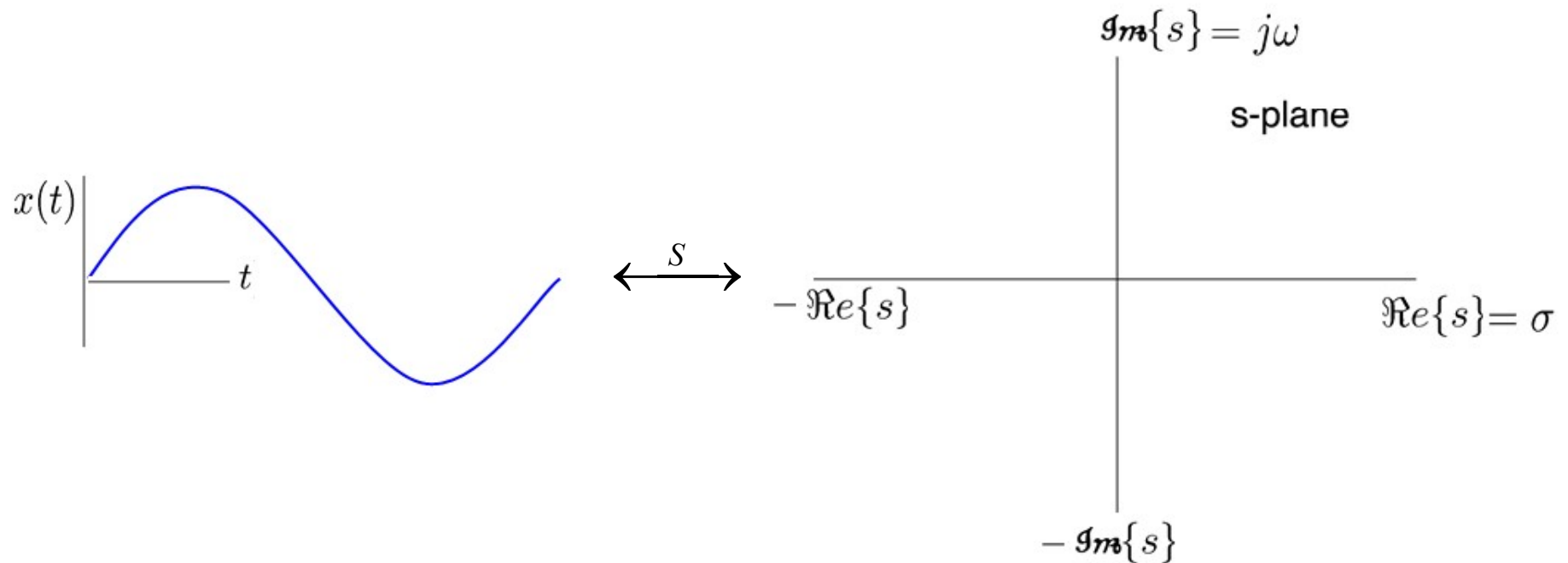
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Laplace (L) Transform

❖ In L-Transform, the continuous time signal $x(t)$ is transformed from time domain into **complex S-plane**.

$$L[x(t)] = X(s) = \int_{-\infty}^{\infty} x(t)e^{-st} dt$$

Where s is complex variable and $s = \sigma + j\omega$



$$\begin{aligned}
 X(s) &= \int_{-\infty}^{\infty} x(t)e^{-st} dt = \int_{-\infty}^{\infty} x(t)e^{-(\sigma+j\omega)t} dt = \int_{-\infty}^{\infty} x(t)e^{-\sigma t} e^{-j\omega t} dt \\
 &= CTFT [x(t)e^{-\sigma t}]
 \end{aligned}$$

❖ L-Transform of a continuous time signal $x(t)$ is denoted by $L[x(t)]$

or

$$x(t) \xleftrightarrow{s} X(s)$$

ROC (Region of Convergence)

$$\text{ROC} = \{s = \sigma + j\omega \text{ so that } \underbrace{\int_{-\infty}^{\infty} |x(t)e^{-\sigma t}| dt}_{\substack{\text{Depends} \\ \text{only on } \sigma \\ \text{not on } \omega}} < \infty\}$$

$$|X(s)| = \int_{-\infty}^{\infty} |x(t)e^{-st}| dt = \int_{-\infty}^{\infty} |x(t)e^{-\sigma t} e^{-j\omega t}| dt$$

$$= \int_{-\infty}^{\infty} |x(t)e^{-\sigma t}| dt < \infty$$

❖ ROC of $X(s)$ is the set of all values of s for which $X(s)$ (Laplace integral) attains finite value

❖ $X(s)$ exists for those value of s for integral converges.

❖ i.e. $X(s)$ exists only for its ROC.

Example 1: Find the L-transform of following signals .

(i) $x[t] = \delta(t)$

(ix) $x[t] = 7e^{-3t}u(t) - 6e^{-2t}u(-t)$

(ii) $x[t] = \delta(t - t_0)$

(x) $x[t] = 7e^{-2t}u(t) - 6e^{-3t}u(-t)$

(iii) $x[t] = \delta(t + t_0)$

(xi) $x[t] = \sin(\omega_0 t)u[t]$

(iv) $x[t] = e^{-at}u[t]$

(xii) $x[t] = \cos(\omega_0 t)u[t]$

(v) $x[t] = -e^{-at}u[-t]$

(xiii) $x[t] = b^{|t|}, b > 0$

(vi) $x[t] = 7e^{-3t}u[t] - 6e^{-2t}u[t]$

(vii) $x[t] = -7e^{-3t}u[-t] - 6e^{-2t}u[-t]$

(viii) $x[t] = e^{-at}u(t) - e^{-bt}u(-t)$

(i) $x[t] = \delta(t)$

Sol:-

$$L\{x(t)\} = X(s) = \int_{-\infty}^{\infty} x(t)e^{-st} dt$$

$$X(s) = L\{\delta(t)\} = \int_{-\infty}^{\infty} \delta(t)e^{-st} dt = 1$$

ROC of $X(s)$ is all s-plane

(ii) $x[t] = \delta(t - t_0)$

$$X(s) = L\{\delta(t - t_0)\} = \int_{-\infty}^{\infty} \delta(t - t_0)e^{-st} dt = e^{-st_0}$$

ROC of $X(s)$ is all s-plane except $\sigma \neq \pm\infty$

(iv) $x[t] = e^{-at}u[t] \longrightarrow$ Causal signal

Sol:-

$$L\{x(t)\} = X(s) = \int_{-\infty}^{\infty} x(t)e^{-st} dt$$

$$L\{e^{-at}u(t)\} = X(s) = \int_{-\infty}^{\infty} e^{-at}u(t)e^{-st} dt$$

$$= \int_0^{\infty} e^{-at} e^{-st} dt = \int_0^{\infty} e^{-(s+a)t} dt$$

$$= -\frac{1}{s+a} \left[e^{-(s+a)t} \right]_0^{\infty} = \frac{1}{s+a}$$

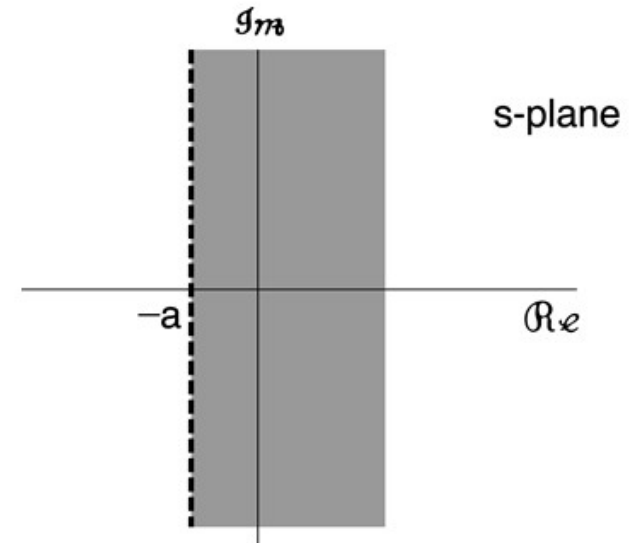
$$\operatorname{Re}(s+a) > 0 \Rightarrow \operatorname{Re}(s) > -\operatorname{Re}(a)$$

or

When ROC of $X(s)$:

$$\sigma + a > 0 \Rightarrow \sigma > -a$$

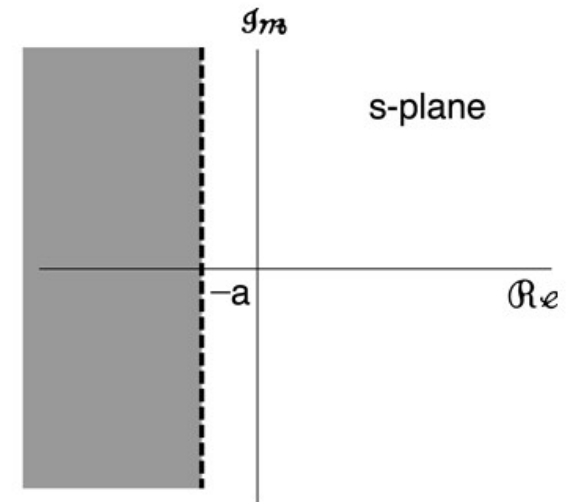
ROC of causal signal is right side of vertical line $\sigma = -a$



(v) $x[t] = -e^{-at}u[-t] \longrightarrow$ Anti causal signal

Sol:-

$$\begin{aligned} X(s) &= L\{-e^{-at}u(-t)\} = \int_{-\infty}^{\infty} -e^{-at}u(-t)e^{-st}dt \\ &= \int_{-\infty}^0 -e^{-at}e^{-st}dt = \int_{-\infty}^0 -e^{-(s+a)t}dt \\ &= \frac{1}{s+a} \left[e^{-(s+a)t} \right]_{-\infty}^0 = \frac{1}{s+a} \end{aligned}$$



$$\text{Re}(s+a) < 0 \Rightarrow \text{Re}(s) < -\text{Re}(a)$$

When ROC of X(s):

or

$$\sigma + a < 0 \Rightarrow \sigma < -a$$

ROC of anti causal signal is left side of vertical line $\sigma = -a$

$$(vi) \quad x[t] = 7e^{-3t}u[t] - 6e^{-2t}u[t]$$

Sol:-

$$7e^{-3t}u(t) \xleftrightarrow{s} \frac{7}{s+3},, \quad \text{ROC: } \sigma > -3$$

$$6e^{-2t}u(t) \xleftrightarrow{s} \frac{6}{s+2},, \quad \text{ROC: } \sigma > -2$$

$$\text{So } 7e^{-3t}u(t) - 6e^{-2t}u(t) \xleftrightarrow{s} \frac{7}{s+3} - \frac{6}{s+2}$$

$$\text{ROC: } (\sigma > -3) \cap (\sigma > -2) = (\sigma > -2)$$

ROC of X(s) is right side of right most vertical line

$$\text{(vii)} \quad x[t] = -7e^{-3t}u[-t] - 6e^{-2t}u[-t]$$

$$-7e^{-3t}u(-t) \xleftrightarrow{s} \frac{7}{s+3},, \quad \text{ROC: } \sigma < -3$$

$$-6e^{-2t}u(-t) \xleftrightarrow{s} \frac{6}{s+2},, \quad \text{ROC: } \sigma < -2$$

$$\text{So} \quad -7e^{-3t}u(-t) - 6e^{-2t}u(-t) \xleftrightarrow{s} \frac{7}{s+3} + \frac{6}{s+2}$$

$$\text{ROC: } (\sigma < -3) \cap (\sigma < -2) = (\sigma < -3)$$

ROC of X(s) is left side of left most vertical line

(viii) $x[t] = e^{-at}u(t) - e^{-bt}u(-t) \longrightarrow$ Non causal Signal

$$e^{-at}u(t) \xleftrightarrow{s} \frac{1}{s+a},, \quad \text{ROC: } \sigma > -\text{Re}(a)$$

$$-e^{-bt}u(-t) \xleftrightarrow{s} \frac{1}{s+b},, \quad \text{ROC: } \sigma < -\text{Re}(b)$$

So
$$e^{-at}u(t) - e^{-bt}u(-t) \xleftrightarrow{s} \frac{1}{s+a} + \frac{1}{s+b}$$

(i) If $b < a$

$$\text{ROC: } (\sigma > -\text{Re}(a)) \cap (\sigma < -\text{Re}(b)) = (-\text{Re}(a) < \sigma < -\text{Re}(b))$$

(ii) If $b > a$ ROC: $(\sigma > -\text{Re}(a)) \cap (\sigma < -\text{Re}(b)) = \varnothing$

X(s) does not exist

$$\text{(ix)} \quad x[t] = 7e^{-3t}u(t) - 6e^{-2t}u(-t)$$

$$7e^{-3t}u(t) \xleftrightarrow{s} \frac{7}{s+3},, \quad \text{ROC: } \sigma > -3$$

$$-6e^{-2t}u(-t) \xleftrightarrow{s} \frac{6}{s+2},, \quad \text{ROC: } \sigma < -2$$

$$\text{So} \quad 7e^{-3t}u(t) - 6e^{-2t}u(-t) \xleftrightarrow{s} \frac{7}{s+3} + \frac{6}{s+2}$$

$$\text{ROC: } (\sigma > -3) \cap (\sigma < -2) = (-3 < \sigma < -2)$$

$$\text{(ix)} \quad x[t] = 7e^{-2t}u(t) - 6e^{-3t}u(-t)$$

$$7e^{-2t}u(t) \xleftrightarrow{s} \frac{7}{s+2},, \quad \text{ROC: } \sigma > -2$$

$$-6e^{-3t}u(-t) \xleftrightarrow{s} \frac{6}{s+3},, \quad \text{ROC: } \sigma < -3$$

$$\text{So} \quad 7e^{-2t}u(t) - 6e^{-3t}u(-t) \xleftrightarrow{s} \frac{7}{s+3} + \frac{6}{s+2}$$

$$\text{ROC: } (\sigma > -2) \cap (\sigma < -3) = \phi$$

$X(s)$ does not exist

Thank You